



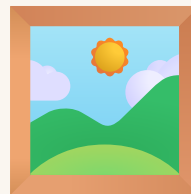
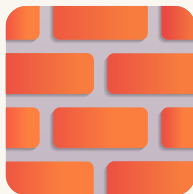
First Steps with (x, y)

Every spot on the wall has a name

벽 위의 모든 칸에는 (x, y) 좌표 이름이 있어요

BIG PICTURE

8 weeks → your own museum 🏛️



It starts with two numbers on a flat wall.

두 개의 숫자에서 미술관까지

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

What is a coordinate ?

A coordinate is a block's **address** — two numbers: (x, y) .

x
how far across →

y
how far up ↑

x = 가로 (옆으로), y = 세로 (위로)

CONCEPT

Counting starts at 0

0

(0, 0)

bottom corner
where counting starts



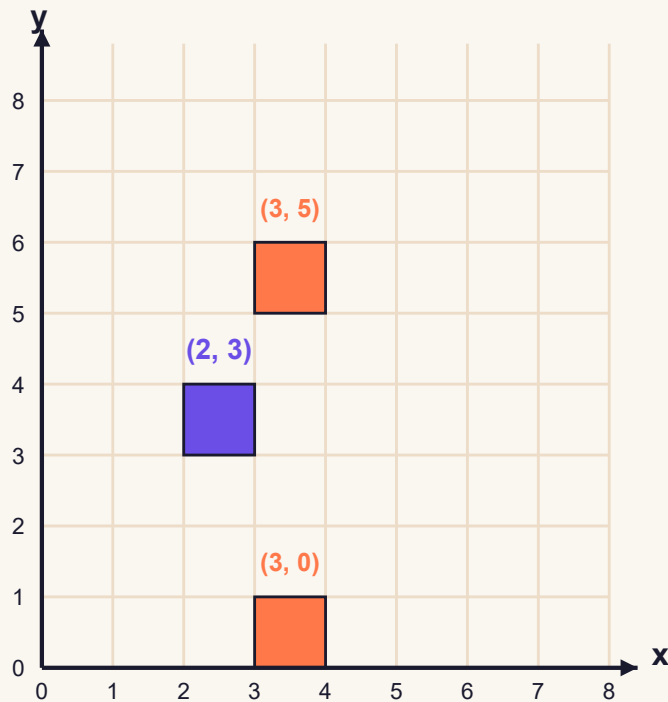
(3, 5)

3 across,
5 up

제일 첫 칸은 1이 아니라 0이에요

SEE IT

The wall is a grid of (x, y)



가로(x)와 세로(y) 두 숫자로 칸의 이름을 정해요

CONCEPT

Same x, different y

```
place gold block at (3, 0)  
place gold block at (3, 5)
```

Both at $x = 3$, so they line up. Only y changed — one is 5 blocks higher.

x가 같으면 한 줄로 서고, y만 바뀌면 위아래로 떨어져요

CODE

Placing a block in Python

```
blocks.place(GOLD_BLOCK, pos(3, 0, 0))  
blocks.place(GOLD_BLOCK, pos(3, 5, 0))
```

The first number is `x`, the second is `y` (height). Same as `(x, y)` on the wall.

첫 번째 = x, 두 번째 = y

CODE

Using a variable for y

```
# move the block up by changing y  
y = 2  
blocks.place(GOLD_BLOCK, pos(0, y, 0))  
y = 6  
blocks.place(GOLD_BLOCK, pos(0, y, 0))
```

Same x, but **y** goes 2 → 6, so the second block lands higher.

PREDICT 🤖

Where do these land?

Predict first — then click reveal

```
place red block at (0, 0)  
place red block at (4, 0)  
place red block at (0, 4)
```

Reveal output 🔍 reveal

VOCAB

New words today

coordinate a block's address, written (x, y)

x how far across (left ↔ right)

y how far up (height)

zero where counting starts: 0, not 1

좌표

가로

세로 / 높이

영 (0)



What's wrong with this code?

Goal: place a block at **(2, 5)** — 2 across, 5 up.

buggy

```
place gold block at (5, 2)
```

Think first — then click next.



DEBUG

The bug

Bug: the numbers are **swapped**. (5, 2) means 5 across and 2 up. x always comes first, y second.

fixed

```
place gold block at (2, 5)
```

YOUR TURN!

Map the wall

- ✓ Open your homework world + Code Builder
- ✓ Place a gold block at `(0, 0)` — your starting corner
- ✓ Pick 3 more coordinates and place a block at each
- ✓ Write down every `(x, y)` you used
- ✓ Send a photo of your wall



QUIZ

QUESTION 1 OF 3

In $(3, 5)$, what does the **3** tell you?

How far up (y).

How far across (x).

How many blocks to place.

Nothing — order doesn't matter.



QUIZ

QUESTION 2 OF 3

Where does counting start on the wall?

$(0, 1)$

Anywhere you like.

$(0, 0)$

$(1, 1)$



QUIZ

QUESTION 3 OF 3

Two blocks share the same x but different y . How are they arranged?

Side by side across the wall.

One directly above the other.

In the same exact spot.

Diagonally.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Wall mapped!

Next: loops to draw whole lines.

다음 주: 반복문으로 선 그리기